

Exact Partition-Refinement Canonicalization and Structural Design Reduction for High-Dimensional Fixed-Effect Models

Functional-dependency certificates before dense dummy materialization

EconHDFE Project

Project Technical Documentation

Implementation reference: `econhdfc/hdfc/structure.py` and `econhdfc/design_structure.py`

September 2026

Abstract

Empirical specifications with high-dimensional fixed effects (HDFEs), categorical factors, and factor interactions often contain exact linear dependencies that are obvious from the partition structure of the categorical variables but expensive to discover after dense dummy expansion. This note develops an exact structural-reduction framework based on partition refinement and functional-dependency certificates. If categorical partition Q refines partition P , then the dummy column space of P is contained in that of Q . This elementary identity yields two complementary operations. First, redundant absorbed fixed-effect partitions can be removed from the numerical HDFE system without changing its column space. Second, categorical and factor-interaction regressor blocks can be reduced before materialization: blocks spanned by absorbed effects can be removed entirely, and a fine full-dummy block can drop one basis column per fully represented coarse cell while preserving the joint span. The same results extend to factor-slope blocks sharing a common continuous monomial. Exact component-level functional dependencies are stored in a directed refinement graph; transitive closure allows certified dependencies among higher-order interactions to be inferred without repeated full-data scans. Sampling is used only to reject false candidates; every data-derived direct edge is certified on the full estimation sample. We prove the column-space identities, establish invariance of the fitted model under the reductions, describe the singleton-pruning recertification rule, and map the results to the `econhdfc` implementation and tests. Partition refinement and functional dependencies are established concepts; the technical contribution is their theorem-backed integration into an HDFE-specific canonicalization and pre-materialization design planner with auditable proof traces.

Formal verification and priority boundary (September 2026). The core mathematical statements in this note are mapped to an independent Lean 4 proof library and have passed complete compilation and per-theorem axiom audit under the assumptions recorded in `formal/mathematical-coverage.md`. This machine-checked status concerns column-space, WLS/FWL, coefficient-selection and stated finite-PPML consequences; it does not certify planner control flow, floating-point behavior, IV/GMM or covariance claims not explicitly mapped, or historical priority. Partition refinement and functional dependencies retain their prior-art status.

Keywords: high-dimensional fixed effects; categorical interactions; design matrices; functional dependencies; partition refinement; collinearity; sparse model compilation.

JEL codes: C01, C13, C63, C87.

Claim boundary. Functional dependencies and partition refinement are established in database theory and structure-aware learning, and ordinary dummy-variable collinearity is textbook linear algebra. The contribution claimed here is the HDFE-specific exact reduction framework: certified removal of redundant absorbed partitions, exact basis reduction of factor/interaction blocks before dense materialization, transitive reuse of component dependencies, and strict separation of numerical canonicalization from requested inference topology.

1 Categorical partitions and refinement

Let a categorical variable induce a partition $P = \{P_1, \dots, P_K\}$ of the estimation rows. Let $D_P \in \{0, 1\}^{N \times K}$ be its full one-hot matrix. For two partitions P and Q , write $Q \preceq P$ when Q is finer than P : every cell of Q is contained in exactly one cell of P .

Definition 1.1 (Refinement map). If $Q \preceq P$, there exists a map f from observed Q levels to observed P levels such that $P(i) = f(Q(i))$ for every estimation row i .

Lemma 1.2 (Dummy-space inclusion). If $Q \preceq P$, then $\text{col}(D_P) \subseteq \text{col}(D_Q)$.

Proof. For each coarse cell p , its indicator equals the sum of the indicators of all fine cells q mapped to p :

$$\mathbf{1}\{P = p\} = \sum_{q: f(q)=p} \mathbf{1}\{Q = q\}.$$

Hence every column of D_P is a linear combination of columns of D_Q . $\square$

Corollary 1.3 (Equivalent partitions). If $P \preceq Q$ and $Q \preceq P$ on the estimation sample, then $\text{col}(D_P) = \text{col}(D_Q)$.

A refinement may be known from the specification itself. For example, the interaction partition `city#year` necessarily refines `year`. Other refinements are empirical functional dependencies: if the estimation sample satisfies `city -> province`, then `city#year` refines `province#year`.

2 Exact canonicalization of absorbed fixed effects

Suppose the HDFE design contains intercept-only blocks $D_{P_1}, \dots, D_{P_G}$. If P_j refines P_i , Lemma 1.2 implies

$$\text{col}([D_{P_1}, \dots, D_{P_i}, \dots, D_{P_j}, \dots, D_{P_G}]) = \text{col}([D_{P_1}, \dots, D_{P_j}, \dots, D_{P_G}]).$$

Therefore P_i may be removed from the *numerical* absorption system without changing the projection space.

Theorem 2.1 (Exact FE partition canonicalization). Let $\mathcal{P}$ be a finite collection of intercept-only categorical FE partitions. Repeatedly remove any partition whose dummy space is contained in another retained partition, using only exact refinement certificates. The resulting maximal collection $\mathcal{P}^*$ satisfies

$$\text{col}(D_{\mathcal{P}}) = \text{col}(D_{\mathcal{P}^*}).$$

Hence every Frisch–Waugh–Lovell residual is unchanged. Coefficients on the unchanged non-FE design are unchanged when its within-transformed columns are identified. In a rank-deficient transformed design, equality of a selected coefficient vector requires the same selection function applied to the same within-transformed design and outcome; using the same minimum-norm rule on differently parameterized full models is not sufficient.

Proof. Each individual removal preserves the concatenated FE column space by Lemma 1.2. Induction over the finite removal sequence gives equality of the initial and final spaces. Orthogonal projection onto equal subspaces is identical, so the within-transformed non-FE variables are unchanged. Under the stated identification or common coefficient-selection condition, the coefficients are unchanged as well. $\square$

The implementation chooses deterministic parents and retains maximal partitions. This is a normalization choice, not part of the theorem.

3 Certification pipeline

A production system should not drop an FE because a small random sample happens to suggest a functional dependency. The implementation therefore uses the following conservative sequence:

1. specification proof when interaction-component containment is sufficient;
2. level-count filter: a finer partition cannot have fewer observed cells than its coarse target;
3. deterministic sample check used only to reject false candidates cheaply;
4. full-data mapping scan for every surviving data-derived direct edge.

Only steps 1 and 4 can certify a removal. The sample check can save work but can never create a proof.

Singleton pruning can change the estimation sample and thereby create a new exact functional dependency. Consequently, if singleton removal changes the row set, canonicalization is rerun on the final estimation sample. Row deletion preserves every previously certified functional dependency: it cannot create a violating pair. Recertification after deletion discovers additional valid reductions and refreshes coding and active-level metadata; it is not needed to rescue a formerly valid dependency. Adding or replacing rows, in contrast, requires fresh certification.

4 Structural regressor reduction before materialization

Consider a full dummy regressor block D_Q and an absorbed FE partition P .

Proposition 4.1 (Block absorbed by a finer FE). *If the absorbed partition P refines regressor partition Q , then $\text{col}(D_Q) \subseteq \text{col}(D_P)$ and the full regressor block contributes no identified direction after absorption. It may be omitted before dense materialization.*

This follows directly from Lemma 1.2.

The reverse relation produces a different exact reduction.

Proposition 4.2 (One fine basis column per represented coarse cell). *Suppose regressor partition Q refines absorbed partition P . In any coarse cell p for which all observed active fine cells are represented, the corresponding absorbed dummy satisfies*

$$\mathbf{1}\{P = p\} = \sum_{q: f(q)=p} \mathbf{1}\{Q = q\}.$$

Therefore, conditional on retaining D_P , one arbitrary active fine dummy in that coarse cell may be removed without changing $\text{col}([D_P, D_Q])$.

Proof. Choose fine level q^* in coarse cell p . Rearranging the identity gives

$$\mathbf{1}\{Q = q^*\} = \mathbf{1}\{P = p\} - \sum_{q \neq q^*, f(q)=p} \mathbf{1}\{Q = q\}.$$

Thus the removed column is in the span of the retained coarse dummy and the other fine dummies. The reverse containment is trivial, proving equality of the joint spans. $\square$

The implementation deterministically removes the last active fine cell in each covered coarse cell, preserving earlier user/level order. Any deterministic choice would be algebraically valid.

5 Common continuous monomials

Let $m \in \mathbb{R}^N$ be a continuous regressor or continuous monomial shared by two categorical interaction blocks. The factor-slope block for partition P is $\text{diag}(m)D_P$.

Lemma 5.1 (Refinement under a shared multiplier). *If $Q \preceq P$, then*

$$\text{col}(\text{diag}(m)D_P) \subseteq \text{col}(\text{diag}(m)D_Q).$$

Proof. Multiply each dummy identity in Lemma 1.2 elementwise by m . The same coefficients express every column of $\text{diag}(m)D_P$ as a linear combination of columns of $\text{diag}(m)D_Q$. $\square$

Proposition 4.2 also carries over after multiplication by the same m , provided the retained coarse columns are multiplied by that same m too. An unmultiplied absorbed dummy does not in general absorb a varying-slope block. This justifies grouping regressor terms by a *continuous signature*: structural reductions are applied only among blocks sharing the same continuous monomial. Reference-coded blocks require care because missing base cells can break the full-sum identity; the implementation therefore tracks active levels explicitly.

6 Component dependency graph

High-order interactions can be expensive to compare directly. Let categorical source components be vertices in a directed graph. Use refinement edge $A \rightsquigarrow B$ when partition B refines partition A . This stores the pair (coarse, fine), opposite to the conventional functional-dependency arrow $B \rightarrow A$. Equivalent partitions can generate directed cycles. The relation is a preorder; only its quotient by mutual determination is a DAG. The implementation computes reachability and does not assume acyclicity.

Lemma 6.1 (Transitivity). *If B refines A and C refines B , then C refines A .*

Proof. Compose the two level maps. $\square$

The planner certifies nearby direct edges first, then maintains transitive closure. If `city` $\rightarrow$ `province` $\rightarrow$ `region`, the `city` $\rightarrow$ `region` relation is inferred without a third full N -row mapping scan.

For interaction partitions, if every categorical component of a coarse interaction is determined by components present in a fine interaction, then the fine joint tuple determines the coarse joint tuple.

Proposition 6.2 (Component-closure interaction certificate). *Let P be an interaction partition generated by components $(A_1, \dots, A_r)$ and Q by $(B_1, \dots, B_s)$. If for every A_j there exists some B_k such that B_k determines A_j through the certified dependency closure (including equality), then Q refines P .*

Proof. A Q cell fixes the values of all B_k . By assumption these values determine every A_j , hence fix the entire coarse tuple $(A_1, \dots, A_r)$. Therefore every Q cell is contained in a single P cell. $\square$

This certificate can prune a high-dimensional interaction block without constructing or rescanning its joint dummy matrix, provided the required component edges have already been exactly certified.

7 Main invariance result

Theorem 7.1 (Structural design reduction preserves the estimable column space). *Consider a regression specification containing absorbed intercept-only categorical FE blocks, categorical factor blocks, and factor-interaction blocks. Apply only the reductions certified by Theorem 2.1, Proposition 4.1, Proposition 4.2, Lemma 5.1, and Proposition 6.2, respecting active/reference-coded levels and shared continuous signatures. Then the span of the combined absorbed-plus-regressor design is unchanged. Consequently, fitted values are unchanged. Estimable linear functions are preserved under the induced coefficient reparameterization; equality of individually named coefficients additionally requires a common identified parameterization and compatible reference restrictions. Equality of column spaces alone does not imply equality of arbitrary coefficient coordinates.*

Proof. Every transformation either removes a block already contained in a retained span or removes a column explicitly represented as a linear combination of retained columns. Therefore each step preserves the total column space. Composition of finitely many span-preserving steps preserves the final span. $\square$

The planner intentionally does not attempt arbitrary collective multi-block rank reasoning. Dependencies not proven structurally fall through to post-absorption numerical rank resolution. This conservative boundary prevents symbolic overreach.

8 Relationship to prior work

Functional dependencies are a standard concept in relational database theory, and structure-aware learning systems have used schema/query dependencies to reduce the cost of statistical learning. In particular, Abo Khamis et al. (2017) develops regression and other learning algorithms that exploit functional dependencies over relational data. HDFE software such as `reghdfe` already accepts factor interactions and performs numerical collinearity handling. The present framework does not claim those ingredients as new.

The HDFE-specific technical contribution is the exact compilation layer joining these ideas: FE column-space canonicalization; sample-certified functional dependencies; basis reduction of full factor and factor-slope interaction blocks before dense materialization; component-graph closure reused across interactions; recertification after singleton pruning; and an auditable proof trace that keeps requested inference topology distinct from the canonical numerical solver topology.

9 Implementation correspondence

Mathematical object	Implementation / validation
FE refinement certification	<code>econhdfe/hdfe/structure.py</code>
FE canonicalization	<code>canonicalize_fixed_effects</code>
Structural term representation	<code>econhdfe/design.structure.py::StructuralTerm</code>
Fine/coarse basis reduction	<code>_coverage_spans_coarse,</code> <code>_drop_one_fine_per_covered_coarse</code>
Component dependency closure	<code>_component_dependency_closure</code>
Structural plan	<code>plan_structural_collinearity</code>
FE canonicalization tests	<code>tests/test_fe_canonicalization.py,</code> <code>tests/test_v07.py</code>
Structural design tests	<code>tests/test_v073_structural_collinearity.py</code>
Dependency-closure tests	<code>tests/test_v074_dependency_omission.py</code>
Stress evidence	<code>benchmarks/bench_structural_collinearity.py,</code> <code>benchmarks/bench_dependency_dag.py</code>

10 Examples

If `city -> province` on the final estimation sample, then

$$\text{city\#year} \preceq \text{province\#year} \preceq \text{year}$$

in the refinement sense (the leftmost partition is finest). Thus an absorbed system requesting

$$\text{firm} + \text{year} + \text{province\#year} + \text{city\#year}$$

can be numerically canonicalized to

$$\text{firm} + \text{city\#year}$$

without changing the FE projection space.

As a regressor example, if `year` is absorbed and a full `quarter\#year` dummy block is requested, then within each year the quarter-year indicators sum to the absorbed year dummy. One quarter-year basis column per fully represented year can be removed before building the dense design.

For a factor-slope example, if x is included as a main effect and the model also requests all Q columns $\mathbf{1}\{q = j\}x$, then

$$x = \sum_{j=1}^Q \mathbf{1}\{q = j\}x.$$

One column can be removed from the later block according to user order while preserving the span.

11 Complexity

A direct refinement check between dense-coded partitions is $O(N + K_{\text{fine}})$ time and $O(K_{\text{fine}})$ memory. The deterministic sample stage is rejection-only and can avoid many full checks in unrelated high-dimensional partitions. The component dependency graph may reduce the number of full scans from

all candidate pairs toward the number needed to establish a sparse hierarchy, because transitive relations are inferred algebraically.

The principal memory benefit of structural design reduction is avoiding materialization of redundant $N \times K$ factor blocks. This can dominate the planner cost for large interactions. Exact speedups are specification dependent and are benchmarked separately from the algebraic claims.

12 Limitations

The current planner is deliberately incomplete. It does not solve arbitrary symbolic rank among several simultaneously dependent blocks; it uses pairwise/refinement identities and sends unresolved dependencies to a numerical Gram-rank resolver after HDFE absorption. Heterogeneous-slope absorbed FE terms are not removed wholesale by the FE canonicalizer because intercept retention can affect numerical conditioning and because their column spaces are not described by pure partition refinement alone. Functional dependencies are sample-specific: row deletion preserves existing dependencies but can create new ones, while adding or replacing rows requires recertification. Specification-derived certificates trust the internally generated component metadata; arbitrary external metadata is not an independent proof. Finally, the framework proves column-space equivalence, not statistical validity of a substantive specification.

13 Conclusion

Exact partition-refinement certificates provide a useful compilation layer between user-facing HDFE specifications and numerical estimation. They allow redundant FE partitions to be removed, factor and interaction blocks to be basis-reduced before materialization, and certified component dependencies to be reused transitively. The resulting model has the same combined column space as the requested design while using a smaller numerical representation. The underlying algebra is elementary; the technical contribution lies in a conservative, auditable HDFE-specific framework that turns the algebra into exact pre-materialization reductions without conflating numerical canonicalization with inference topology.

A Lean formalization of structural design invariance

Scope and assumptions. This appendix records exact real column-space results on a fixed finite sample. The proof library uses Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. Verification is attached to an exact source commit passing the full build and per-theorem dependency audit. It is not a claim about the Python planner.

Refinement and functional dependence. `Refines` is defined by an actual level map on all sample rows. `partitionSpace_eq_span` identifies the categorical coefficient-map range with the full indicator-column span. The dummy-space lemma is `partitionSpace_le`; equivalent partitions are handled by `equivalent_partition_spaces`. The dependency lemmas are `refines_trans`, `refines_restrict` and `interaction_refines`. Restriction to fewer rows preserves a true dependency; the statements do not certify a dependency inferred from a random subsample or an unverified metadata claim.

Canonicalization and absorbed blocks. `canonicalize_one` removes a coarse partition only while its fine witness remains. `absorbed_block_within_zero`, together with the proved refinement inclusion, shows that a regressor block already contained in the absorbed space has no residualized direction. The mathematical within operator is constructed from existence and uniqueness of finite positive-weight WLS; it is not introduced as an unspecified numerical routine.

Active and reference-coded columns. `coarse_active_sum` requires completeness only for realized fine levels inside the coarse cell being reduced. `drop_active_fine` retains the coarse indicator and all other fine columns needed for its sum identity. This permits missing reference cells elsewhere, as well as unused labels, but does not permit a further deletion when an observed fine cell required by the identity is already absent. `drop_one_fine` is the earlier complete-block special case. Reductions in several coarse cells compose by reapplying the same condition on each current active set.

Shared continuous multipliers. `shared_multiplier_le` proves inclusion after multiplication by the same realized continuous vector. `drop_active_fine_multiplier` and `finite_reduction_common_multiplier` cover basis elimination and its finite composition. The multiplier may vanish or be negative; the essential condition is that it acts on both the fine columns and the retained coarse witness. An unmultiplied coarse FE is not silently treated as a slope witness.

Main column-space theorem. `StructuralStep` describes only the preceding mathematical reductions and their common-multiplier and unchanged-context extensions. Its constructors carry level maps and coverage conditions, not an assumed equality of the spaces. `structural_step_space_eq` proves equality for each case, and `structural_trace_space_eq` proves it for a finite sequence. The term “trace” here denotes a finite mathematical proof composition, not a formal model of the software planner’s control flow.

OLS/WLS and coefficient semantics. `wls_fit_exists_unique` proves existence and uniqueness of fitted vectors; `structural_trace_wls_fit` gives fitted-value invariance. `full_wls_implies_within` and `within_wls_implies_full` connect full and within WLS optima. `full_wls_identified_fe_invariance` therefore establishes equality of identified non-FE coefficients when the FE spaces agree and the non-FE design, sample and weights remain fixed. For rank-deficient transformed designs, `within_selection_invariance` requires the same selection function of the same transformed design and outcome. It does not equate minimum-norm coefficients of differently parameterized full models. `estimable_coefficient_fiber` handles coefficient functionals annihilating the design kernel; `estimable_fit_function_invariance` handles a common linear functional of fitted values.

Finite PPML corollary. `ppmlLoss` is the actual weighted Poisson objective, up to constants independent of the predictor. `ppml_midpoint_strict` proves strict midpoint convexity for strictly positive weights, using strict convexity of the exponential and cancellation of the linear response term. `finite_ppml_predictor_unique` proves uniqueness of an attained finite predictor on the affine space determined by the design and offset. `finite_ppml_fitted_invariance` and `finite_ppml_response_invariance` give equal η and $\exp(\eta)$ under equal design spaces, unchanged sample, weights, response and offset. `structural_trace_ppml` connects this to the actual allowed mathematical reductions. Existence of a finite optimum is deliberately not asserted; separation and numerical convergence are different questions.

Boundaries. No IV/GMM moment or covariance theorem is inferred from column-space equality alone. Reference-restriction rules outside the stated witness conditions, planner decisions, sampled certification, sparse data structures, numerical ranks and floating-point behavior are not part of the mathematical claim. The Lean results neither certify software implementations nor establish historical novelty.

References

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